

Tide retrospective: *global constancy of the harmonic Gibbs covariance*

Daniel Murfet (with Claude Opus 4.7)

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Abstract

A 428-line Tide-loop excursion in the `laplace` seabed strengthening Tide 13's local-at- $h=0$ constancy of the harmonic Gibbs covariance to a *global* closed-form identity: $\text{Cov}_h(x, x) = 1/(\lambda t)$ for all $h \in \mathbb{R}$. The proof avoids the `GibbsObservable` bundle entirely (Tide 13 needed three such hypotheses) by direct closed-form evaluation of the three perturbed integrals Z_h , $\langle x \rangle_h$, $\langle x^2 \rangle_h$, each via square completion plus translation-invariance of Lebesgue (`MeasureTheory.integral_add_right_eq_self`). Lives at `Laplace/OneD/HarmonicCovGlobalConstancy.lean` on branch¹ off `main` (commit 141997c). Zero `sorry`, zero `axiom`, zero `native_decide`; build clean. Step 2 deliberation with GPT-5.5 Pro converged on *Candidate B'* on the first round; Step 3 was mechanical Lean bookkeeping with three small idiom frictions and no thrashing-rescue consult.

1 Setting

The Susceptibility Primer's Section 4 develops the harmonic case as the leading-order specialisation of the Laplace expansion. Tide 5 proved κ_3 vanishes for the linear observable (x, x, x) at the harmonic potential; Tide 10 supplied the `GibbsRegularity` instance for that setup; Tide 13 composed those into the cross-susceptibility derivative theorem $\partial_h \text{Cov}_h(x, x)|_{h=0} = -t \cdot \kappa_3 = 0$, recording the global-constancy form as a future tide candidate in the retrospective Follow-ups:

GPT-5.5 Pro: Global-constancy strengthening (the runner-up). Prove `Threepoint.gibbsCov (volume) ((λ/2).²) (id) t h (id) (id) = 1/(λt) forall h, not just locally constant at h = 0`. *Direct via shifted — Gaussian translation invariance plus the closed — form partition already in Tide 10. Avoids GibbsObservable hypotheses entirely.*

This Tide implements that runner-up. The candidate qualifies as a Tide step under the standard heuristic: closed-form scalar answer, single statement, reachable from existing seabed without new infrastructure. Tide-pick on 7 May ranked it the top recommendation.

The Step 2 deliberation² proposed three target shapes:

Candidate A. One headline equation for Cov_h .

Candidate B. Two named perturbed-expectation lemmas ($\langle x \rangle_h = -h/\lambda$ and $\langle x^2 \rangle_h = 1/(\lambda t) + h^2/\lambda^2$), with Cov_h as a 5-line corollary.

¹Branch: `tide/global-constancy-harmonic-cov` in `laplace`, commit 4aca7a8.

²Tide log: `projects/primer/tide-log/2026-05-07-tide-global-constancy-harmonic-cov.md`. GPT-5.5 Pro consult preserved at the sibling `gpt55-tide-global-constancy-harmonic-cov.v1.md`.

Candidate C. An abstract *translation-of-gibbsExp* lemma for arbitrary integrable observables.

GPT recommended a hybrid: Candidate B promoted to *Candidate B'* by adding a standalone `partitionFunction_h_harmonic` lemma on top, since the closed form $Z_h = e^{th^2/(2\lambda)} \sqrt{2\pi/(\lambda t)}$ makes the denominator non-vanishing immediate, packages the square-completion step cleanly, and is reusable for free-energy / log-partition arguments downstream. Both parties voted B'. The numerical sanity check (six (λ, t, h) cases, agreement to $\sim 10^{-14}$ relative) confirmed the closed forms before the proof touched Lean.

2 The thing formalised

For the harmonic Gibbs measure $\mu_{t,h}(x) \propto \exp(-t((\lambda/2)x^2 + hx))$ on $\mathbb{R}$ against Lebesgue, with $\lambda, t > 0$ and any $h \in \mathbb{R}$,

$$\text{Cov}_h(x, x) = \frac{1}{\lambda t}.$$

The covariance is independent of the perturbation h outright; not just at the basepoint.

The Lean signature, in namespace `Laplace.OneD`:

```
theorem gibbsCov_h_id_id_harmonic_eq
  {lam t : ℝ} (hlam : 0 < lam) (ht : 0 < t) (h : ℝ) :
  Threepoint.gibbsCov (volume : Measure ℝ)
    (fun x : ℝ => lam / 2 * x ^ 2)
    (fun x : ℝ => x) t h
    (fun x : ℝ => x) (fun x : ℝ => x)
  = 1 / (lam * t)
```

The hypothesis list is just $\lambda, t > 0$ — no `GibbsObservable` bundle. This is a strict strengthening of Tide 13: the latter required three `GibbsObservable` hypotheses for x, x^2, x^3 to discharge differentiation under the integral, and delivered only the value of the derivative. The square-completion route bypasses both.

The headline theorem is a 5-line corollary of two named perturbed expectations, also exported:

```
theorem gibbsExp_h_id_harmonic_eq ... = -h / lam
theorem gibbsExp_h_sq_harmonic_eq ... = 1 / (lam * t) + h ^ 2 / lam ^ 2
```

together with the standalone

```
theorem partitionFunction_h_harmonic ... =
  Real.exp (t * h^2 / (2 * lam)) * Real.sqrt (2 * Real.pi / (lam * t))
```

3 Proof strategy

The mathematical kernel is the pointwise square-completion identity

$$\frac{\lambda}{2}x^2 + hx = \frac{\lambda}{2}\left(x + \frac{h}{\lambda}\right)^2 - \frac{h^2}{2\lambda},$$

which makes the perturbed Boltzmann factor a translate of the unperturbed one, modulo the constant $e^{th^2/(2\lambda)}$ that cancels in every expectation ratio:

$$e^{-tL_h(x)} = e^{th^2/(2\lambda)} \cdot e^{-t(\lambda/2)(x+h/\lambda)^2}.$$

The substitution $y = x + h/\lambda$ then maps the perturbed integrand onto the unperturbed one. Lebesgue measure on $\mathbb{R}$ is translation-invariant, so by Mathlib's `MeasureTheory.integral.add_right_eq_self` the substitution preserves integrals over the whole line.

This collapses the three perturbed numerators (and the partition function Z_h) to integrals against $e^{-t(\lambda/2)y^2}$ of polynomial expressions in $y-h/\lambda$. Linearity of integration plus parity (Tide 10's `harmonic_int_pow_odd`) plus the explicit second moment (Tide 10's `harmonic_int_pow_even` at $k = 1$) closes the four integrals:

$$\begin{aligned} Z_h &= e^K Z_0, & \int x e^{-tL_h} &= -\frac{h}{\lambda} \cdot e^K Z_0, \\ \int x^2 e^{-tL_h} &= \left(\frac{1}{\lambda t} + \frac{h^2}{\lambda^2}\right) \cdot e^K Z_0, \end{aligned}$$

where $K := th^2/(2\lambda)$ and $Z_0 := \sqrt{2\pi/(\lambda t)}$. The e^K factor cancels: $\langle x \rangle_h = -h/\lambda$, $\langle x^2 \rangle_h = 1/(\lambda t) + h^2/\lambda^2$, and $\text{Cov}_h = \langle x^2 \rangle_h - \langle x \rangle_h^2 = 1/(\lambda t)$.

The Lean file factors this into an internal helper `integral_with_perturbation_eq_shifted`, which takes any $f : \mathbb{R} \rightarrow \mathbb{R}$ and rewrites $\int f(x) e^{-tL_h(x)} dx = e^K \int f(y - h/\lambda) e^{-tL_0(y)} dy$. Each public theorem then specialises to $f \in \{1, x, x^2\}$. The final Cov_h theorem is a 5-line algebraic corollary of the two perturbed-expectation lemmas via `field.simp; ring`.

The integrability witnesses for 1 , y , $y \cdot y$ against the unperturbed Gaussian are routed through Mathlib's `integrable_exp_neg_mul_sq`, `integrable_mul_exp_neg_mul_sq`, and `integrable_rpow_mul_exp_neg_mul_sq` (with $s = 2$), with small `convert` bridges to map $\exp(-(\lambda t/2) \cdot y^2)$ to the seabed convention $\exp(-(t \cdot (\lambda/2)y^2))$.

4 Roadblocks and resolutions

The mathematical content matched the Step 2 plan exactly, so the frictions in Step 3 were all Lean idiom and all small. Three concrete items:

Mathlib lemma name miscalls. The first draft used `integrable_id_mul_exp_neg_mul_sq` and `integrable_pow_mul_exp_neg_mul_sq` as integrability witnesses for $y \cdot e^{-by^2}$ and $y^2 \cdot e^{-by^2}$. Neither exists. Mathlib has `integrable_mul_exp_neg_mul_sq` for the first; for the second, the existing `laplace` pattern (used in `Laplace/OneD/Quartic.lean` and elsewhere) is to take `integrable_rpow_mul_exp_neg_mul_sq` at $s = 2$ and convert via `Real.rpow_natCast`. Recording in `CLAUDE.md` for the next session.

Double-factorial cast normalisation. Tide 10's `harmonic_int_pow_even` returns the second moment in the form

$$((2k - 1)!! : \mathbb{R}) \cdot \sqrt{2\pi} \cdot (\lambda t)^{-(\lceil k \rceil + 1/2)}.$$

For $k = 1$ this should reduce to $1 \cdot \sqrt{2\pi} \cdot (\lambda t)^{-3/2}$, but the factor $((1)!! : \mathbb{R})$ does not reduce by `rfl` alone — `Nat.doubleFactorial 1 = 1` is definitional, but the cast $((1 : \mathbb{N}) : \mathbb{R}) = (1 : \mathbb{R})$ is the `Nat.cast_one` lemma, not `rfl`. The fix:

```

have hdf : (((2 * 1 - 1 : )!! : ℝ)) = 1 := by
  change (((1 : ) : ℝ)) = 1
  exact Nat.cast_one

```

The two `change` steps are definitional ($2 \cdot 1 - 1 = 1$ in $\mathbb{N}$, then $1!! = 1$ from the definition's case $| 1 \Rightarrow 1$), so `change` discharges them silently; only `Nat.cast_one` is needed at the end. A common micro-pitfall on first encounter; worth promoting to `CLAUDE.md`.

Pi.add vs single-lambda for `integral_sub + integral_add`. The expanded second-moment integrand is $(y-c)^2g(y) = y^2g - 2cyg + c^2g$, integrated as $(\int y^2g) - (\int 2cyg) + (\int c^2g)$. To recombine into a single integral for an `integral_congr_ae` step, both `integral_sub` and `integral_add` must fire in reverse. The trap is *order*: applying $[\leftarrow \text{integral_add}, \leftarrow \text{integral_sub}]$ fails because the outer $+$ has two *integrals* on either side, not an integral on one side and a sum-integrand on the other. The correct order is $[\leftarrow \text{integral_sub} \dots, \leftarrow \text{integral_add} \dots]$: the inner $-$ combines first (yielding the single integrand $y^2g - 2cyg$), and then the outer $+$ combines that with the third term. The `CLAUDE.md` note about `Pi.add` vs single-lambda is adjacent — type-ascribing the integrability witness to a single lambda gives Lean the right shape for the rewrite to fire — but the order issue is one step earlier and has its own gotcha.

There were no thrashing-rescue GPT consults. The Step 2 deliberation gave the right plan; Step 3 was three friction points each fixed with a single edit.

5 What was Mathlib, what was new

Mathlib provided. The integrability witnesses (`integrable_exp_neg_mul_sq`, `integrable_mul_exp_neg_mul_sq`, `integrable_rpow_mul_exp_neg_mul_sq`); the translation-invariance lemma `MeasureTheory.integral_add_right_eq_self` (additive form of `integral_mul_right_eq_self` via `to_additive`, firing on Lebesgue volume as an additive Haar measure); the constant-multiplication `integral_const_mul`; the sub/add-decomposition `integral_sub / integral_add`; the rpow algebra `Real.rpow_neg_one`, `Real.rpow_add`, `Real.rpow_natCast`, `Real.sqrt_eq_rpow`, `Real.sqrt_mul`; and `integral_congr_ae` for the pointwise rewrite.

Laplace/Threepoint provided. Tide 10's `partitionFunction_harmonic`, `harmonic_int_pow_even` (at $k = 1$), and `harmonic_int_pow_odd` (at $k = 0$). Threepoint's definitional `gibbsExp` and `gibbsCov`.

What was new. The square-completion identity packaged as `harmonic_h_square_completion` and `harmonic_h_exp_factor`; the generic shift-of-integrand helper `integral_with_perturbation_eq_shifted` (taking f as a parameter, applying square completion + translation invariance once); three integrability bridges (`integrable_harmonic_gauss`, `integrable_harmonic_gauss_id`, `integrable_harmonic_gauss_sq`) converting between the Mathlib convention $\exp(-b \cdot y^2)$ and the seabed convention $\exp(-(t \cdot (\lambda/2) \cdot y^2))$; three closed-form bridges (`harmonic_int_zero/first/second_pow...`) re-expressing Tide 10's integrals without the `harmonicPotential / Laplace.partitionFunction` wrappers; and the four public theorems on top.

The novelty is in the *packaging*, not the mathematics. The square-completion proof is textbook physics; what was new was finding the right Mathlib lemma surface to land it in ~ 100 lines of Lean rather than ~ 400 .

6 Lessons

- *Avoiding `GibbsObservable` bundles is sometimes the better route.* Tide 13 carried three `GibbsObservable` hypotheses to differentiate the covariance under the integral; the square-completion route here makes the same theorem stronger (global, not just basepoint) *and* hypothesis-free. The generalisation: when a perturbed Gaussian admits a closed-form reduction to the unperturbed one, prefer evaluating the integral directly to differentiating it.
- *Two-pass cast normalisation for double factorials.* The pattern `change ((n : ) : ℝ) = ...; exact Nat.cast_one` (or `Nat.cast_succ`, etc.) handles the “definitional-on-Nat-but-not-rfl-on-real” case cleanly. Worth promoting to the laplace `CLAUDE.md` alongside the existing `Pi.add` note.
- *For `integral_add/integral_sub` reverse rewrite chains, order matters.* Combine inner operations before outer ones. The `CLAUDE.md` `Pi.add-vs-single-lambda` note covers the integrand-shape side; this run adds the combine-order side.
- *Tide-shape candidates with prior GPT-deliberation rarely thrash.* GPT-5.5 Pro had already named this candidate at the Tide 13 retrospective deliberation; the v1 consult here was largely confirmation plus the $B \rightarrow B'$ refinement. Step 3 was mechanical. This is a data point in favour of writing follow-up candidates in retrospectives *with their proof outline*, because that work doesn’t have to happen again at Step 2 of the next tide.
- *The `shift-of-integrand` helper is a candidate for promotion.* `integral_with_perturbation_eq_shifted` generalises beyond the harmonic case: any L admitting a square-completion shift would benefit. Defer until a second use.

7 Follow-ups

- $\langle x^k \rangle_h$ for general $k \in \mathbb{N}$. The same shift method gives $\langle x^k \rangle_h = \mathbb{E}_0[(y - h/\lambda)^k]$, which expands by the binomial formula into combinations of unperturbed even moments. A natural follow-up tide (~ 300 lines together for $k \in \{1, \dots, 4\}$). Pairs with G2 / G5 in the candidates survey.
- *Discharge Tide 13’s `GibbsObservable` hypotheses.* G4 (already landed) closed three of the six required hypotheses for monomial observables; the remaining three are mixed-power products $x \cdot x^2, x^2 \cdot x, x \cdot x \cdot x^2$ and similar. The global-constancy theorem here makes Tide 13’s conditional theorem redundant if these are discharged: $\partial_h \text{Cov} = 0$ follows from $\text{Cov} = \text{const}$. Worth recording but not prioritising.
- *The structural measure-equality framing (Candidate C).* GPT noted: at the normalised probability-measure level, $\mu_h = (x \mapsto x - h/\lambda)_* \mu_0$, no scalar factor. This is the right framing for higher cumulants and would unify several perturbed-Gaussian theorems. Defer until a second use materialises; in Lean, building `Measure.map + withDensity` infrastructure costs more than the shift identity it proves.
- *\leanref-tag the primer.* The primer’s §3 has the centroid-shift remark $\langle x \rangle_h = -h/\lambda$ and the variance-invariance remark $\text{Var}_h = 1/(\lambda t)$. Pin `\laplaceCommit = 4aca7a8` and tag both with `\leanref`. Ten-minute edit on a paper-side branch. Pairs with K2 in the survey.

- *Promote `integral_with_perturbation_eq_shifted` on second use.* Currently a private helper. Second use likely from the $\langle x^k \rangle_h$ tide above; at that point lift to a named public theorem, possibly to `lean-common` on third use.

Acknowledgements

GPT-5.5 Pro contributed the deliberation-stage advisory at Step 2, including the $B \rightarrow B'$ refinement (promoting `partitionFunction_h_harmonic` to a standalone export) and the API guidance toward `integral_add_right_eq_self` over `MeasurePreserving.integral_comp`. The full consult is preserved at `projects/primer/tide-log/gpt55_tide_global_constancy_harmonic_cov_v1.md`.

This Tide branched off the post-G4 `main` (commit 141997c) without depending on G4 directly: G4 discharged the `GibbsObservable` hypotheses needed for Tide 13's κ_3 -route, while this Tide bypasses those hypotheses entirely. The two routes are independent paths to the same neighbourhood of the harmonic Gibbs theory; landing both gives the primer's harmonic case both an unconditional cross-susceptibility result and a stronger global-constancy result.